A learning image keeps the input origins of an internal response

Real input → selected ViT response → positive contributions at the original input locations



Input image

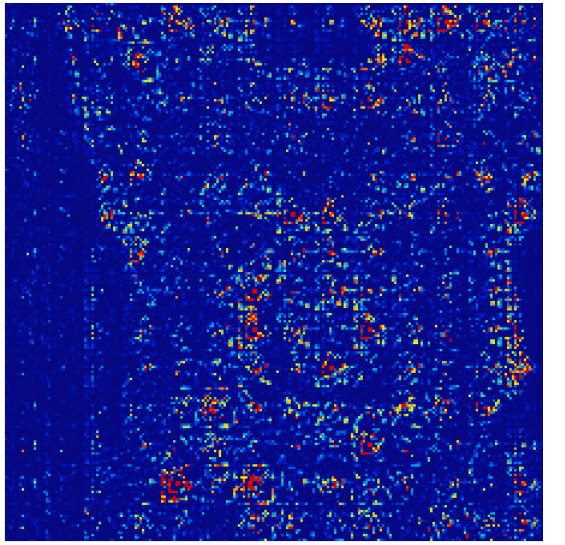
Trained ViT · Block 0

Patch embedding → Attention

Channel 255 · 196 patch tokens

$$s(x) = \sum_{t=1}^{196} O_{t,255}(x)$$

Response change Δs = 10.2007



Positive learning image · 224 × 224

low high

Positive part · P99 = 0.0204 · deep blue = zero/low

$$L_s(i) = (x_i - x_{0i}) \int_0^1 \partial_i s(x_0 + \alpha(x - x_0))\, \mathrm{d}\alpha$$

Main panel shows max(Ls,0) · full signed map retained in supplement · no spatial smoothing

Full signed map sum = 10.2007

Displayed positive mass = 99.7551